

Fast Digitization in XR of Reality-Linked Gaussian-Splatted Proxies

Benjamin Yang*

Shanying Liu*

Kirthana Natarajan*

Steven K. Feiner*

Department of Computer Science
Columbia University

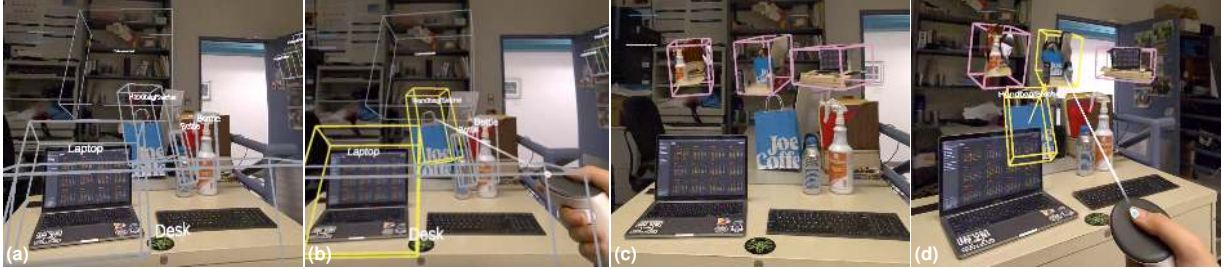


Figure 1: Gaussian-Splatted Proxies pipeline. (a) Real-world objects detected via Quest 3 passthrough camera. (b) User selects objects of interest, a laptop and bag, with controller ray and button press. (c) Feed-forward 3D Gaussian splatting reconstructs the selected objects. (d) Extracted objects become interactive 3D proxies that users can grab, manipulate, and use to reference real-world objects in XR. User points at the generated bag proxy to highlight the bounding box of its real-world counterpart.

ABSTRACT

Capturing real objects as interactive 3D assets in XR typically requires offline photogrammetry. We present Gaussian-Splatted Proxies, a system for fast object digitization in XR using feed-forward 3D Gaussian splatting. With a Quest 3 headset, users select objects detected by the system through the passthrough cameras, then capture images that an edge server reconstructs as interactive Gaussian splats. Our system extracts individual objects from scene reconstructions without any per-scene training by projecting detection bounding boxes into 3D space. The resulting assets can be grabbed, inspected, rotated, and used to select the corresponding objects in the physical world. <https://youtu.be/orx3gFPBNpA>

Index Terms: Human–computer interaction (HCI)–Interaction paradigms–Mixed/augmented reality; Computing methodologies–Computer graphics–Image manipulation–Image-based rendering.

1 INTRODUCTION

Extended Reality (XR) applications increasingly require 3D content derived from the real world. Traditional photogrammetry and structured-light scanning rely on offline processing. Recent advances in 3D Gaussian Splatting (3DGS) [5] have produced photorealistic novel view synthesis; however, most methods, including some commercial tools, require per-scene optimization that takes minutes to hours.

Polycam (<https://poly.cam>) and Meta Hyperscape Capture [8] enable 3D Gaussian capture from mobile devices, but they rely on cloud processing with per-scene optimization over many images, which is too slow for interactive XR workflows. Gracia (<https://gracia.ai>) and Meta Hyperscape Capture render pre-captured high-resolution Gaussian splats in XR at high frame rates, but Gracia does not support in-headset capture and both do not support processing on a locally-owned machine. We present **Gaussian-Splatted Proxies**, which combine *feed-forward* 3DGS with on-device object detection from just two images in seconds

to produce photorealistic reconstructions within seconds, all without leaving XR. Feed-forward methods like Splatt3R [13] perform stereo matching without requiring camera pose calibration.

Our system combines on-device object detection with server-side Splatt3R reconstruction as shown in Figure 1. Objects are detected through a Meta Quest 3 passthrough camera, after which the user selects targets of interest, and receives interactive 3D Gaussian splat reconstructions within seconds. We introduce zero-shot semantic object extraction by combining YOLO11 (<https://docs.ultralytics.com/models/yolo11>) detection bounding boxes with Gaussian filtering to isolate individual objects from reconstructions without any per-scene training. This allows users to capture and extract multiple objects in a scene. Thus, we contribute:

- An XR pipeline for fast object digitization using feed-forward 3DGS without leaving XR.
- Zero-shot semantic object extraction by projecting detection boxes into 3D space to isolate individual objects.
- Integration of captured objects as interactive XR assets that can be grabbed, inspected, manipulated, and used to reference their corresponding real-world objects.

2 RELATED WORK

3DGS has recently been explored in XR. Kari and Abtahi [4] use pre-captured 3DGS to create virtual–physical decoupling illusions. VR-GS [3] supports physics-aware editing of Gaussian splats, while SCGS [10] applies semantic segmentation for immersive scene editing. 3DGS has been evaluated as an alternative to video passthrough for static environments [11]. However, these approaches require per-scene optimization, limiting their applicability for fast capture.

Feed-forward methods eliminate per-scene training. Splatt3R [13] reconstructs 3DGS from stereo image pairs using a feed-forward network built on MAST3R [6] stereo matching, producing results in under a second. We use Splatt3R and extend it with semantic object extraction, enabling users to capture, isolate, and interact with individual objects.

3 IMPLEMENTATION

We developed Gaussian-Splatted Proxies in Unity 6000.0.62f1 using a Meta Quest 3 headset. We rely on an edge server driven by

*e-mail: {benyang, sl5678, kmn2161, feiner}@columbia.edu

a Windows 11 computer with an Intel® Core™ i9-11900K processor and Nvidia GeForce RTX 4090 graphics card. Our architecture has three stages: object detection and selection, feed-forward reconstruction, and semantic object extraction, as shown in Figure 2.

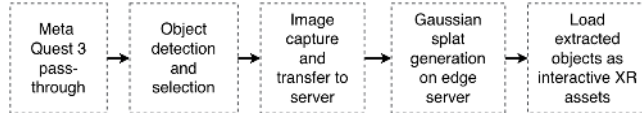


Figure 2: System architecture.

3.1 Object Detection and Selection

The Quest 3 passthrough cameras provide a live view of the physical environment, as shown in Figure 1(a). Object detection runs on-device using YOLO11 trained on the Objects365 dataset [12] via the Unity Sentic inference engine, overlaying bounding boxes on detected objects. Users aim their controller ray at a target object and confirm selection with a button press, highlighting the object with a colored outline, as shown in Figure 1(b).

Upon selection, the system captures two images separated by a brief delay as the user moves their head slightly. This head motion provides the viewpoint difference required for stereo reconstruction. A floating alignment guide helps maintain stability during capture, changing color to indicate readiness.

3.2 Feed-Forward Reconstruction

The two captured images are transmitted to an edge server running Splatt3R [13], which performs feed-forward 3DGS reconstruction in approximately 0.25 seconds at 512×512 resolution. Splatt3R is built on MAST3R [6] for stereo matching and camera pose estimation, eliminating the 30–50 minutes of per-scene optimization required by traditional methods. The reconstructed PLY file is streamed back to the Quest 3, where it is rendered untethered using custom Gaussian Splatting shaders optimized for mobile GPU. Reconstructed objects spawn at a fixed height of approximately 1 meter (convenient for grab interaction), with metric scale preserved from reconstruction and colliders automatically sized to the splat bounds, as shown in Figure 1(c). End-to-end latency from capture to interaction is typically 2–4 seconds, dominated by network transfer and server inference.

3.3 Semantic Object Extraction

To extract individual objects from scene reconstructions, we project detection bounding boxes into 3D in real time using the Quest 3 depth-sensing API, with box depth estimated from visible dimensions. Each 3D bounding box is represented as an oriented bounding box. We transform each splat’s position into the box’s local coordinate frame and retain only those that fall within the box dimensions. This produces a filtered PLY file containing only the target object’s splats. The zero-shot approach requires no per-scene training or segmentation models. Extracted objects are registered as Unity components, with colliders sized to match the splat bounds, allowing users to pick up, inspect, and reposition the digitized objects. Each proxy maintains a reference to its source detection, enabling users to point at a proxy to highlight its real-world counterpart’s bounding box, as shown in Figure 1(d).

4 CONCLUSIONS, LIMITATIONS, AND FUTURE WORK

We presented Gaussian-Splatted Proxies, an XR system for fast object digitization using feed-forward 3DGS with zero-shot semantic extraction. Our pipeline allows users to reconstruct and interact with photorealistic 3D representations of physical objects.

Current limitations include dependency on object texture (textureless or reflective surfaces produce sparse Gaussians), single image-pair processing without multi-view fusion (which remains

an open research challenge for feed-forward methods), processing time, bounding box projection using single-point depth sampling that works best for upright objects on horizontal surfaces, and reduced extraction accuracy for objects that are too close to other objects or are partially occluded.

We envision several applications for this workflow. In *remote collaboration*, users could share photorealistic 3D representations of physical objects with remote collaborators who can inspect them from any angle, extending prior work on 3D referencing [9] and world-stabilized annotations [1]. In *design review*, captured physical prototypes could be overlaid on CAD models in XR to identify discrepancies. Our captured objects can also be used to select their corresponding real-world objects [9, 7]. *Robot training* is a promising application in which captured objects could serve as photorealistic digital twins for simulation-based imitation learning [2]. Moving forward, we plan to implement multi-view Gaussian fusion to improve reconstruction completeness and evaluate the system in collaborative robotics scenarios.

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REFERENCES

- [1] S. Gauglitz, B. Nuernberger, M. Turk, and T. Höllerer. World-stabilized annotations and virtual scene navigation for remote collaboration. In *Proc. UIST*, 2014. doi: 10.1145/2642918.2647372 2
- [2] X. Han, M. Liu, Y. Chen, J. Yu, X. Lyu, Y. Tian, B. Wang, W. Zhang, and J. Pang. Re3Sim: Generating high-fidelity simulation data via 3D-photorealistic real-to-sim for robotic manipulation. *arXiv preprint arXiv:2502.08645*, 2025. doi: 10.48550/arXiv.2502.08645 2
- [3] Y. Jiang, C. Yu, T. Xie, X. Li, Y. Feng, H. Wang, M. Li, H. Lau, F. Gao, Y. Yang, and C. Jiang. VR-GS: A physical dynamics-aware interactive Gaussian splatting system in virtual reality. In *ACM SIGGRAPH 2024*, 2024. doi: 10.1145/3641519.3657419 1
- [4] M. Kari and P. Abtahi. Reality promises: Virtual-physical decoupling illusions in MR via invisible mobile robots. In *Proc. UIST*, 2025. doi: 10.1145/3746059.3747660 1
- [5] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis. 3D Gaussian splatting for real-time radiance field rendering. *ACM ToG*, 42(4), 2023. doi: 10.1145/3592433 1
- [6] V. Leroy, Y. Cabon, and J. Revaud. Grounding image matching in 3D with MAST3R. In *Proc. ECCV*, 2024. doi: 10.1007/978-3-031-73220-1_5 1, 2
- [7] X. Liu, D. Jia, X. C. Liu, M. Gonzalez-Franco, and Z.-T. Chen. Reality proxy: Fluid interactions with real-world objects in MR via abstract representations. In *Proc. UIST*, 2025. doi: 10.1145/3746059.3747709 2
- [8] Meta. Meta horizon hyperscape capture. <https://www.meta.com/experiences/meta-horizon-hyperscape-capture-beta/8798130056953686/>, 2025. Accessed: 2026-01-06. 1
- [9] O. Oda and S. Feiner. 3D referencing techniques for physical objects in shared augmented reality. In *Proc. ISMAR*, 2012. doi: 10.1109/ISMAR.2012.6402558 2
- [10] H. Schieber, J. Young, T. Langlotz, S. Zollmann, and D. Roth. Semantics-controlled Gaussian splatting for outdoor scene reconstruction and rendering in virtual reality. In *Proc. IEEE VR*, 2025. doi: 10.1109/VR59515.2025.00056 1
- [11] A. Schleising, C. Kunert, T. Schwandt, and W. Broll. Virtual pass-through: Evaluating 3D Gaussian Splatting as an alternative to conventional video pass-through in static environments. In *Proc. ISMAR*, 2025. doi: 10.1109/ISMAR67309.2025.00131 1
- [12] S. Shao, Z. Li, T. Zhang, C. Peng, G. Yu, X. Zhang, J. Li, and J. Sun. Objects365: A large-scale, high-quality dataset for object detection. In *Proc. ICCV*, 2019. doi: 10.1109/ICCV.2019.00852 2
- [13] B. Smart, C. Zheng, I. Laina, and V. A. Prisacariu. Splatt3R: Zero-shot Gaussian splatting from uncalibrated image pairs. *arXiv preprint arXiv:2408.13912*, 2024. doi: 10.48550/arXiv.2408.13912 1, 2